Supplementary material

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Description. Implementation details, historical analyses, negative results, clinical-standards context, subgroup diagnostics, and a comparison of representative cuffless blood-pressure studies.

Primary claims and conclusions are reported in the main manuscript.

Appendix A
Supplementary Implementation Details

A. Multi-Wavelength Pulse Processing

For each device, visible and infrared pulse rates are estimated around the scheduled reference time using a window of up to 12 s, with at least 7.5 s and 150 samples required. Timestamps are converted to seconds, sorted, and deduplicated; signals are interpolated to a uniform grid using the median interval, with inferred sampling rate clipped to 20–500 Hz. A third-order 0.55–4 Hz Butterworth filter is applied forward and backward after detrending. Welch spectra use up to 8 s segments (at least 256 samples when available), with 50% overlap. The dominant frequency is sought in 0.65–3.5 Hz; spectral concentration c is the fraction of pulse-band power within 0.18 Hz of that peak.

An autocorrelation pulse estimate searches lags corresponding to 39–210 BPM. Let a be the selected autocorrelation peak clipped to $[0, 1]$, and h_F, h_A the spectral and autocorrelation rates. When $|h_F - h_A| \leq 15$ BPM, the channel estimate is $(ch_F + ah_A)/(c + a)$; otherwise the estimate with the larger reliability component is used. Channel quality is

$$q = \mathrm{clip}\big(\sqrt{ca}\, g, 10^{-4}, 1\big)\,, \tag{1}$$

$$g = \begin{cases} 1, & |h_F - h_A| \leq 15, \\ e^{-|h_F - h_A|/30}, & \text{otherwise.} \end{cases} \tag{2}$$

The tone-blind fusion estimate is $(q_V h_V + q_I h_I)/(q_V + q_I)$. The optional tone-dependent arm replaces the weights by $q_V e^{-\alpha z}$ and $q_I e^{\alpha z}$, with $z = \mathrm{clip}[(\mathrm{MST} - 1)/9, 0, 1]$. Candidate α values are $\{0, \pm 0.25, \pm 0.5, \pm 0.75, \pm 1, \pm 1.5, \pm 2, \pm 3\}$. Four participant folds within each outer training set score these fixed candidate rules. The selected nonzero exponent must improve mean fold MAE by at least 0.10 BPM and improve every inner fold; otherwise $\alpha = 0$. The seed is 20260827, with repeat/fold offsets defined in the companion script. These methods estimate pulse rate, not BP.

B. VitalDB Extraction and Model Selection

Eligible cases have all three required tracks and a case duration of at least 1,800 s. Cases are deterministically shuffled with seed 20260827; positions 1–60 define development and 61–300 define the candidate holdout. The saved audit contains 55 development cases/subjects and 228 eligible holdout cases from 227 subjects, with no subject overlap. Eligibility and sampling are retrospective, and selecting windows across the completed record is not an online scheduling policy.

Candidate centers start no earlier than 120 s, proceed in 30-s increments, and require at least 900 s of usable overlapping record. PPG is retained at approximately 100 Hz. A 20-s window must contain at least 75% of the expected samples and 85% finite values; gaps are interpolated. Values are clipped to the median plus/minus eight robust SD estimates, detrended, and filtered with a third-order 0.5–8 Hz bandpass (upper cutoff limited by sampling rate). Pulse quality is spectral concentration multiplied by $\exp[-\min(\text{interval CV}, 2)]$; windows below 0.04 are excluded. Pulse frequency search, spectral concentration, and beat-interval limits follow the companion extraction code. This quality filter is held fixed, but a no-quality-filter ablation is not reported.

Reference BP is the median of at least three plausible values within ± 5 s of the center. SBP must lie in 70–220 mmHg and DBP in 35–130 mmHg; local robust SD must not exceed 8 and 6 mmHg, respectively. Pulse pressure must lie in 20–130 mmHg. These reference-based exclusions do not use prediction errors but restrict the evaluated BP distribution. Cases require at least eight valid windows. The first three define median BP and feature anchors; up to 24 later windows are selected using evenly spaced indices over the valid future windows.

The context model uses anchor BP, $\log(1 +$ elapsed seconds), age, sex, and preoperative hypertension (missing hypertension is coded as zero). The PPG model adds current and calibration-relative optical features listed by the columns function in the companion script. Candidate estimators comprise standardized ridge regression with 13 logarithmically spaced penalties from 10^{-3} to 10^3 and three absolute-loss gradient-boosting configurations. Development-only five-fold participant validation selects estimators, followed by a fixed blend weight from $\{0, 0.25, 0.5, 0.75, 1\}$; a nonzero weight requires at least 0.10 mmHg improvement over context. Development selection is not independently nested across the estimator and blend choices, but holdout observations are excluded from both. Delta predictions are clipped at training-target 1st/99th percentiles.

Both endpoints selected ridge context models with penalty approximately 31.6228. The SBP PPG model uses absolute-loss boosting, learning rate 0.03, 15 leaves, regularization 5, 250 iterations, and minimum leaf size 20; the DBP PPG model uses ridge with penalty approximately 31.6228. Both selected a constant 0.5 PPG weight. The saved predictions satisfy this fixed-blend identity to numerical precision. Errors are averaged over all windows from all cases of a subject before subject bootstrap; the subject contributing two cases remains one bootstrap unit.

C. Audit Scope and Reproduction Status

The independent-dataset saved-result audit uses 10,000 participant bootstrap resamples with seed 20260910 for the pulse comparisons and verifies window uniqueness, case/subject counts, development/holdout separation, the fixed VitalDB blend, and the pulse-fusion equation. The AuroraBP analysis was separately rerun: the context arms and all three calibration doses were refitted across three repeats of five participant folds. The package records hashed feature inputs, software versions, selected features and gates, row-level outer-fold predictions, and paired participant contrasts. The historical subgroup model was also refitted. The rerun starts from feature banks; raw-waveform extraction is outside its scope.

Appendix B
Historical Development and Auxiliary Analyses

The following analyses preserve the development record. They are secondary or exploratory. The final subgroup subsection reports newly refitted estimates; development codes C0–C4 refer to the historical arms.

A. Exploratory Endpoint-Specific Calibration and Warning Router

Table S2 reports the historical post-primary endpoint analysis retained from the development record; it was not refitted in the V38 primary rerun. A nested, tightly clipped residual calibrator reduced SBP participant-level MAE from 7.611 to 7.555 mmHg, a paired improvement of 0.056 mmHg (95% CI 0.020–0.091; 55.9% of participants helped). The absolute gain is small, and tail behavior changed little (P90 absolute error 16.862 → 16.820 mmHg; P95 22.184 → 22.120 mmHg). DBP candidate specialization did not improve inner-fold performance consistently, so the base prediction was retained at 5.784 mmHg. This endpoint-specific policy is more conservative than adopting one post-processing rule for both targets. The exploratory DBP router was therefore frozen as warning-only. Extreme DBP change prevalence was 20.5% (95% CI 19.1–21.9%). The router achieved ROC AUC 0.794 (0.778–0.809), average precision 0.517 (0.483–0.555), and Brier score 0.130 (0.122–0.137). At the nested operating threshold, warning precision was 0.773 (0.695–0.854), but recall was only 0.043 (0.035–0.052) and the warning rate was 1.1% (0.9–1.4%). The router is thus a rare high-precision flag at the evaluated threshold; it is not a screen for most extreme changes, does not reduce reported DBP MAE, and must not be interpreted as a clinical safety mechanism.

B. Historical Duplicate-Reading Noise Diagnostic

Table S3 retains a historical duplicate-reading diagnostic, not a universal achievable-error bound. Estimating per-reading noise as $\sigma_{P-S}/\sqrt{2}$ assumes independent, equal-variance observer errors. The supplied code distinguishes a stored average of two readings from a single-reading target and propagates target and anchor errors for a legacy two-time-point difference. Its 1.754/2.412 mmHg Gaussian absolute-noise estimates and 2.6%/9.2% variance ratios therefore must not be treated as validated floors for the primary three-reading anchor. Shared observer error, physiological change during measurement, and the actual target-averaging rule can change the estimates. A primary-endpoint noise calculation would require those components and their covariances; subtracting a putative floor from model MAE does not measure recoverable physiological error.

C. Recalibration Cadence

The historical code evaluated nominal recalibration intervals of 1, 3, 6, 12, and 24 h, with nearly identical MAE (8.593–8.594 mmHg SBP; 6.020–6.022 mmHg DBP). Those labels do not establish distinct effective recalibration schedules in the analyzed short laboratory sessions. The stored comparison lacks an event-count and elapsed-time audit sufficient to interpret a cadence effect. We therefore withdraw the inference that cadence beyond one hour is immaterial and make no recommendation about recalibration frequency from this analysis.

D. Multi-Window Feature Averaging

Averaging attractor features over multiple 10 s windows per recording (up to 3 available) was hypothesized to reduce measurement noise (Table S4 shows within/between SD ratios up to approximately 0.45). It did not help: holdout MAE with multi-window averaging (6.393/4.828 mmHg SBP/DBP) was statistically indistinguishable from, and nominally slightly worse than, a matched single-window configuration (6.346/4.803 mmHg), both close to the legacy single-window baseline (6.315/4.789 mmHg). We report this as a negative result (Table S1).

E. Historical Fixed-Split Session-Anchored Ladder

During earlier development, the model-development ladder was evaluated on a fixed 501/168 participant split. Table S5 preserves that analysis for transparency: C4 reduced MAE from 8.856/6.582 to 7.147/5.579 mmHg. Because the same 168 participants were examined across successive ablations, these values are historical development evidence and are not the paper's primary validation result. The corresponding repeated participant-validation

TABLE S1
Interventions Tested and Not Adopted (Elimination Log)

Intervention	Times tested	Outcome
Absolute-error loss (vs. squared)	3	Never CI-supported: +0.152 [−0.001, 0.311] (51.8% helped); +0.102 [−0.028, 0.240]; +0.163 [−0.008, 0.345] (exactly 50.0% helped).
HCA feature augmentation	2	No CI-supported benefit: −0.077 [−0.183, 0.024]; +0.043 [−0.026, 0.115] (53.0% helped).
Signal-level pigmentation (Beer–Lambert) correction	2	First variant increased error; the repeated-CV corrected channel changed MAE by −0.007 [−0.042, 0.029] SBP and −0.030 [−0.062, 0.003] DBP – no established overall benefit (the main manuscript's repeated-CV pigmentation-correction subsection).
Demographic (Fitzpatrick-band) rebalancing / group gating	5	Consistently harmful: e.g., -0.291 $[-0.420, -0.170]$ mmHg SBP; widens between-band gap in every run (the main manuscript's Secondary Pigmentation-Intervention Audit).
Multi-window feature averaging	1	No measurable benefit; nominally worse (Section B-D).
Nested gate/threshold search (early pipeline)	1	Higher error than a plainer reference baseline (8.253/8.183 vs. 8.109 mmHg SBP).
Passed sanity check: metadata-only permutation control	1	AUC = 0.492 (≈ chance) – is consistent with chance under this permutation control; does not exclude other confounds.

TABLE S2
Exploratory Endpoint-Specific Nested Analysis (653 Participants)

Target	Arm	MAE	P90 AE	P95 AE
SBP	Base	7.611	16.862	22.184
SBP	Clipped residual	7.555	16.820	22.120
DBP	Base	5.784	12.043	15.630
DBP	Selected policy	5.784	12.043	15.630
SBP paired gain: 0.056 mmHg [0.020, 0.091]; DBP gain: 0.000.				

TABLE S3
Historical Duplicate-Reading Diagnostic (8,918/8,916 SBP/DBP pairs)

Quantity (mmHg)	SBP	DBP
σ(primary − secondary reading)	3.108	4.275
σ_{cuff} (per-reading noise)	2.198	3.023
Legacy Gaussian absolute-noise estimate	1.754	2.412
Observed SD of Δ	13.70	9.98
Legacy noise/observed variance ratio	2.6%	9.2%
Residual SD under legacy assumptions	13.52	9.51
Early reference model MAE	8.11	–

TABLE S4
Historical Within- and Between-Recording Feature SD

Feature	Within SD	Between SD	SD ratio
σ_m	0.0046	0.0102	0.449
CSI	0.0043	0.0138	0.311
Recurrence rate	0.0040	0.0130	0.305
Kurtosis	0.0902	0.3086	0.292
RQA entropy	0.0734	0.2520	0.291
Skewness	0.0386	0.1389	0.278
Sample entropy	0.0133	0.0527	0.252
Determinism	0.0114	0.0475	0.240
Spectral entropy	0.0092	0.0488	0.189
Permutation entropy	0.0028	0.0190	0.150

TABLE S5
Historical Fixed-Split Session-Anchored Ladder (168 Evaluation Participants)

Arm	SBP MAE [95% CI]	DBP MAE [95% CI]
Calibration retention	8.856	6.582
C0 legacy (squared)	7.956 [7.424, 8.501]	6.120 [5.641, 6.647]
C1 absolute loss	7.793 [7.270, 8.316]	6.014 [5.513, 6.546]
C2 +context	7.156 [6.740, 7.618]	5.651 [5.186, 6.146]
C3 +context+HCA	7.113 [6.703, 7.550]	5.636 [5.178, 6.126]
C4 +context (squared)	7.147 [6.736, 7.593]	5.579 [5.174, 6.004]
Reduction, calib. → C4	1.709 (19.3%)	1.003 (15.2%)

estimates in the main manuscript's primary validation table are modestly worse (7.523/5.743 mmHg for protocol-enriched context), consistent with mild optimism from the fixed split rather than a collapse of the effect.

F. Calibration-Row Leakage Audit

Table S6 reports all four checks from the main manuscript's calibration-row leakage audit. Check A shows calibration-flagged rows are smaller in mean $|\Delta|$ than unflagged rows (6.774 vs. 9.902 mmHg SBP, development set) but not trivially near-zero, and the fraction of rows within 1 mmHg is nearly identical between flagged and unflagged rows (9.8% vs. 8.6%). The decisive Check D – flag removed from features, flagged rows excluded from both fit and scoring – yields a context gain of +0.614 mmHg SBP $[0.346, 0.889]$ and +0.513 mmHg DBP $[0.268, 0.743]$, both CI-supported and close to the unaudited historical figures. Leave-one-out attribution (Figure S2) shows larger predictive contributions from recording duration and post-exercise activity, not the calibration flag (LOO damage for the flag: 0.0000 SBP / 0.0027 DBP). One honest qualification: the single largest contributor to the context gain is recording duration (0.244 mmHg SBP / 0.366 mmHg DBP of leave-one-out damage), which is an acquisition property (likely proxying protocol step or signal stability) rather than a purely physiological state variable. "Context = posture and activity" is only partly right; post-exercise activity (0.154/0.144 mmHg) is the clearest genuinely physiological

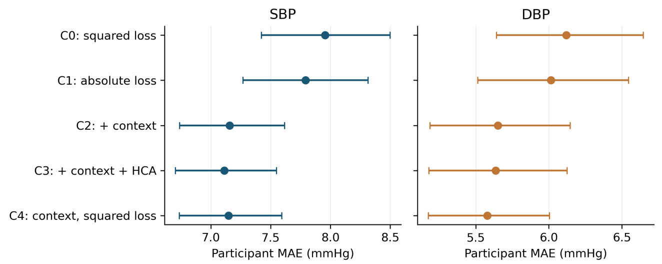

Fig. S1. Historical fixed-split session-anchored ladder with 95% participant-bootstrap CIs. This figure documents model development; the main manuscript's primary validation table provides the primary repeated participant-validation result.

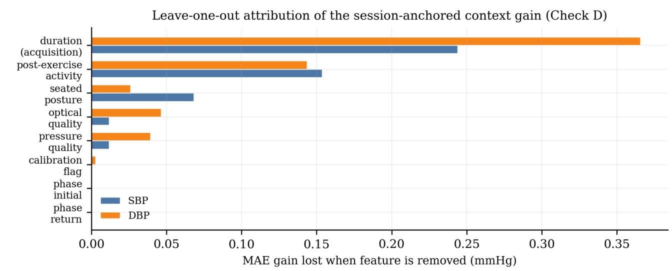

Fig. S2. Leave-one-out attribution of the session-anchored context gain (Check D). The largest contributor is recording duration – an acquisition property, not a purely physiological one – followed by post-exercise activity.

TABLE S6
Calibration-Row Leakage Audit (168 holdout participants)

Check	SBP gain [CI]	DBP gain [CI]	Verdict
B (flagged excl.)	0.741 [0.440, 1.062]	0.606 [0.372, 0.854]	Supp.
C (flag dropped)	0.809 [0.501, 1.122]	0.538 [0.309, 0.770]	Supp.
D (strict)	0.614 [0.346, 0.889]	0.513 [0.268, 0.743]	Supp.

TABLE S7
BHS Grading and AAMI/ISO 81060-2 Criterion 1
(Session-Anchored C4 Model, Measurement-Level)

Target/Arm	≤5mm	≤10mm	≤15mm	Grade	AAMI-1
SBP calibration	45.9%	72.3%	83.9%	D	Fail
SBP baseline (C0)	45.8%	73.8%	86.6%	C	Fail
SBP model (C4)	48.0%	75.1%	89.7%	C	Fail
DBP calibration	56.0%	82.9%	92.2%	B	Fail
DBP baseline (C0)	56.0%	83.1%	93.6%	B	Fail
DBP model (C4)	58.7%	85.6%	94.9%	B	Pass

contributor, with posture third.

G. Clinical Standards Context

Table S7 reports BHS grading and AAMI/ISO 81060-2 Criterion 1 for the C4 session-anchored model. The model improves cumulative within-5/10/15 mmHg percentages over both calibration retention and the legacy baseline for both SBP and DBP, and reaches BHS Grade B for DBP (Grade C for SBP); it passes AAMI Criterion 1 for DBP (mean error 0.645, SD $7.992 \leq 8$ mmHg) but not for SBP (SD $9.654 > 8$ mmHg). We report both gradings as indicative context, not a validation claim: BHS and AAMI/ISO were written for cuff devices measuring absolute BP against an independent reference, and this model is calibrated per participant.

H. What Did Not Work

Table S1 summarizes every intervention tested and rejected in this study, alongside one passed sanity check. We regard this table as carrying as much evidentiary weight as Table S5: a result that survives repeated, adversarially-minded attempts to break it is more credible than one presented without them.

I. Refitted Fixed-Estimator and Primary-Cohort Subgroups

The historical prior-anchored estimator was refitted under the recorded environment, and Table S8 reports its newly generated predictions. These estimates supersede the earlier rounded subgroup values; they do not assert bitwise recovery of the original run. The companion change record retains the old-value comparison. The same estimator is applied within each analysis before predictions are grouped by Fitzpatrick band. The primary wearable-state summary uses the independently regenerated repeated-CV predictions, not the historical fixed split. These are exploratory marginal intervals. Differences in sample size, anchoring, and validation design prevent direct comparison between the two analysis blocks. In particular, the small V–VI samples do not establish parity or equivalence, irrespective of the ordering of the point estimates.

References

David Dasa and Philip Davies. Skin tone and diagnostic equity in contactless blood pressure screening: A prospective observational field evaluation of remote photoplethysmography in nigeria. BMJ Open, 16(6): e119311, 2026. doi: 10.1136/bmjopen-2026-119311.

Sergio González, Wan-Ting Hsieh, and Trista Pei-Chun Chen. A benchmark for machine-learning based non-invasive blood pressure estimation using photoplethysmogram. Scientific Data, 10:149, 2023. doi: 10.1038/s41597-023-02020-6.

Jingon Joung, Chul-Woo Jung, Hyung-Chul Lee, Moon-Jung Chae, Hae-Sung Kim, Jonghun Park, Won-Yong Shin, Changhyun Kim, Minhyung Lee, and Changwoo Choi. Continuous cuffless blood pressure monitoring using photoplethysmography-based PPG2BP-net for high intrasubject blood pressure variations. Scientific Reports, 13:8605, 2023. doi: 10.1038/s41598-023-35492-y.

Timothy Oladunni and Farouk Ganiyu Adewumi. Attractor-vascular coupling theory: Formal grounding and empirical validation for AAMI-standard cuffless blood pressure estimation from smartphone photoplethysmography. arXiv preprint arXiv:2605.10871, 2026. doi: 10.48550/arXiv.2605.10871.

TABLE S8
Regenerated Participant MAE by Fitzpatrick Band

Analysis	Target	Band (n)	MAE	95% CI
Prior refit	SBP	I–II (133)	5.837	[5.513, 6.178]
Prior refit	SBP	III–IV (29)	5.568	[4.703, 6.549]
Prior refit	SBP	V–VI (6)	5.341	[3.847, 6.841]
Prior refit	DBP	I–II (133)	4.439	[4.118, 4.795]
Prior refit	DBP	III–IV (29)	4.781	[3.674, 6.370]
Prior refit	DBP	V–VI (6)	4.020	[2.931, 5.197]
Primary CV	DBP	I–II (516)	6.078	[5.805, 6.362]
Primary CV	DBP	III–IV (112)	6.435	[5.862, 7.045]
Primary CV	DBP	V–VI (25)	6.053	[5.065, 7.122]
Primary CV	SBP	I–II (516)	7.680	[7.355, 8.022]
Primary CV	SBP	III–IV (112)	7.786	[7.050, 8.610]
Primary CV	SBP	V–VI (25)	7.646	[6.052, 9.471]

TABLE S9
Representative Cuffless BP Studies

Method	Dataset (N)	Split	Calibration	SBP	DBP	Skin-tone report
PPG2BP-Net (Joung et al., 2023)	Intraoperative (4,185 participants)	Participant-independent	Required	$0.209{\pm}7.509^{*}$	$0.150{\pm}4.549^{*}$	Not reported
Calibration-free CV-dynamics (Samimi and Dajani, 2023)	UCI/MIMIC-II (225 participants)	Participant-independent	None	7.41	3.32	Not reported
Benchmark baseline (González et al., 2023)	Four public PPG–BP datasets	Participant-level CV	Mixed	15.60 (Sensors)	–	Not reported; quantifies leakage inflation
AVCT companion work (Oladunni and Adewumi, 2026)	Smartphone PPG	Single-point calibration	1 reading	2.05	1.67	Not reported
rPPG equity field study (Dasa and Davies, 2026)	Nigeria, Fitzpatrick V/VI (306 enrolled; 249 analyzed)	Independent cuff reference	None	15.4	10.9	Fitzpatrick V/VI
Wearable-state model (this study)	AuroraBP (653 participants)	3×5-fold participant CV	3 readings	7.697 [7.407, 8.008]	6.139 [5.899, 6.391]	Fitzpatrick I–VI; V–VI underpowered
Protocol-enriched upper bound	AuroraBP (653 participants)	Same internal CV	3 readings	7.523 [7.243, 7.815]	5.743 [5.527, 5.977]	Protocol metadata limit transportability
Context-plus-PPG holdout	VitalDB (227 held-out participants)	Locked participant holdout	3 windows	15.375	8.833	Not available; temporal test only

* Mean error ± SD rather than MAE; not directly comparable with the other rows.

Hamed Samimi and Hilmi R. Dajani. A PPG-based calibration-free cuffless blood pressure estimation method using cardiovascular dynamics. Sensors, 23(8): 4145, 2023. doi: 10.3390/s23084145.